

TASK 1 – Temperature Monitoring and LCD Display using Arduino UNO

OBJECTIVE

To monitor temperature using DHT22 sensor and display the readings on LCD using Arduino UNO.

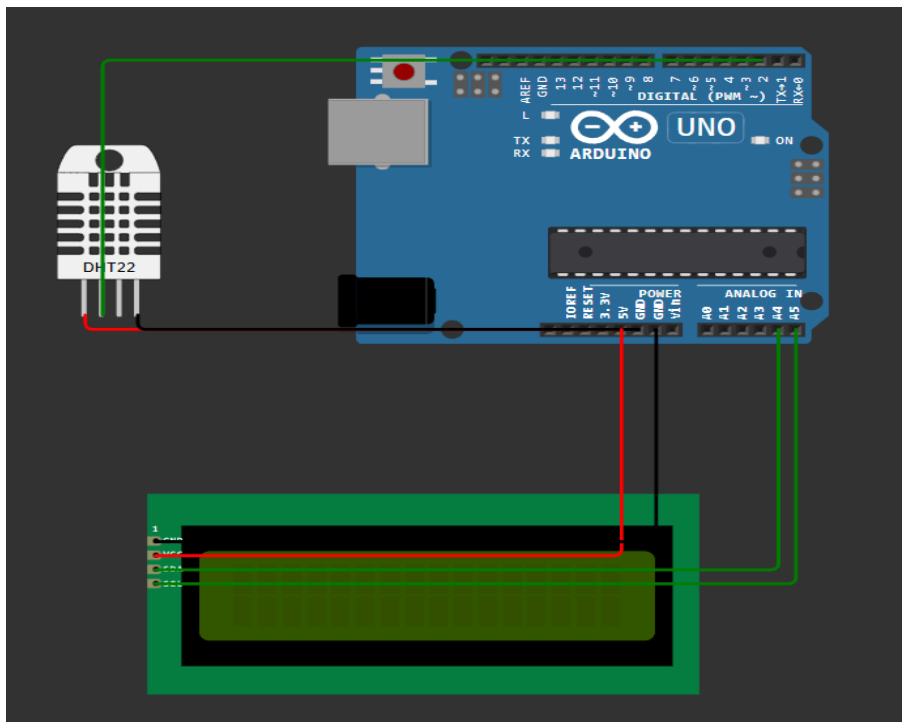
COMPONENTS USED

- Arduino UNO
- DHT22 Sensor
- 16x2 I2C LCD
- Jumper Wires

SOFTWARE USED

- Wokwi Simulator
- Arduino IDE

CIRCUIT DIAGRAM



ARDUINO CODE:-

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>
#define DHTPIN 2
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);
void setup() {
  Serial.begin(9600);
```

```

dht.begin();

lcd.init();
lcd.backlight();

lcd.setCursor(0,0);
lcd.print("Temperature");
}

void loop() {

    float temp = dht.readTemperature();

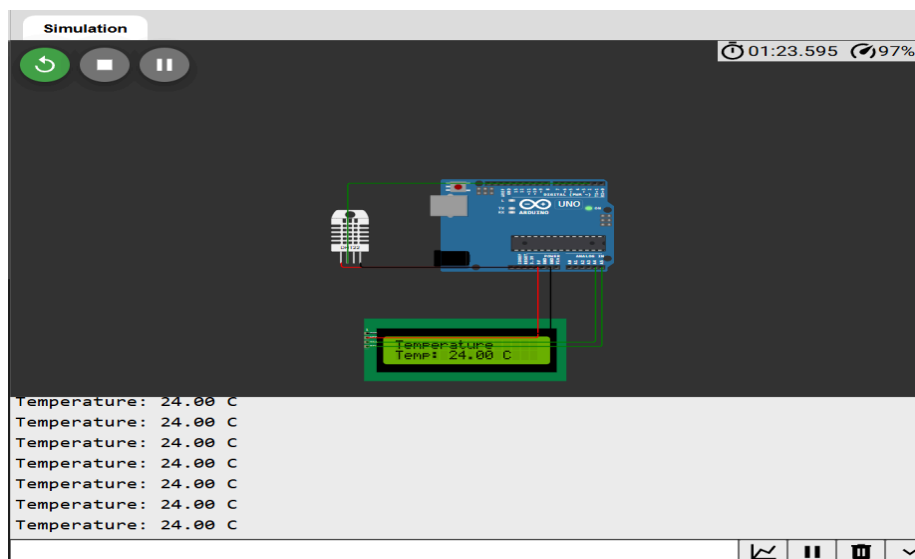
    Serial.print("Temperature: ");
    Serial.print(temp);
    Serial.println(" C");

    lcd.setCursor(0,1);
    lcd.print("Temp: ");
    lcd.print(temp);
    lcd.print(" C ");

    delay(1000);
}

```

OUTPUT :-



CONCLUSION

Temperature readings were successfully displayed on LCD and Serial Monitor.